

mesoestetic®

.com

85



Me

tricology
strength and vitality
from the root

contains
exosomes

the element for skincare experts

hair care solutions

Alopecia is a condition that affects both men and women, and it consists in abnormal hair loss. It is due to multifactorial causes and can occur at various stages of life.

tricology controls the different types of hair loss through a comprehensive, multimodal approach based on the main alopecia pathways. With repairing and revitalizing action, it acts on the scalp to stop hair loss, promote hair growth, lengthen the life cycle of hair and strengthen it from the root.



Multimodal approach

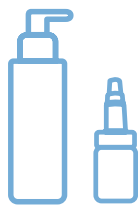
Combining the professional method with an associated home regimen is essential to achieving the target result.

transepidermal solution



 treatment at the clinic

topical home care



 home regimen

oral nutraceutical



 home regimen



Comprehensive
treatment:
professional +
home-based

Differential formulas:
innovation and
efficiency

Agreeable functional
and pharmaceutical
formats

Scientific studies
that support the
results

Types of alopecia

target scope with the trichology range

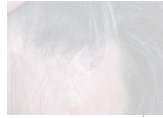
Reactive hair loss

seasonal



- Autumn and spring
- Stress (↑ cortisol)
- Nutritional deficit

concomitant



- Scarring from burns or accidents
- Traumatic
- Some diseases
- Medicinal (chemotherapy)

postpartum¹



Telogen effluvium:
from 60-75 days after
delivery until the end of
breast-feeding

Progressive hair loss

menopause²



- Hormonal imbalances
- Water and mineral deficiency
- Melanin deficiency

male androgenic



80%
prevalence³

female androgenic



50%
prevalence³

autoimmune



areata
frontal fibrosis

alopecias with a
hormonal component

Caucasian ancestry³

Based on the main pathways of androgenic alopecia

Pathways of alopecia

Hormonal pathway
5α-reductase

Hyperseborrhoea
5α-reductase

Inflammation

Vascularization

Metabolic deficiency

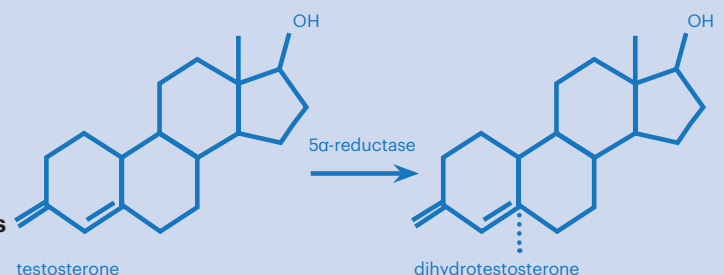
Actions of the trichology range

- ⊘ Blocks testosterone in the follicle, thereby lengthening the hair life cycle
- ⋏ Excessive sebum, which causes follicular death due to blockage
- ⋏ Local microinflammation
- ⋏ Blood microcirculation and oxygenation of the scalp
- ⋏ Supply of vitamins, peptides, growth factors and nutrients

What is 5α-reductase?

An enzyme that transforms testosterone into dihydrotestosterone (DHT), which has a devastating effect on hair follicle hair cycles.

Inhibiting this enzyme normalizes hair cycles and delays the onset of alopecia.



hair care solutions

Cutting-edge formulas with repairing and revitalizing action that act on the scalp to stop hair loss, thereby lengthening the hair's life cycle and strengthening it from the root.



Formula innovation

Differentiating and highly effective active substances that act on the analogous pathways of the main therapeutic targets in the treatment of alopecia

Formulas with plant exosomes*

contains **exosomes**

Small vesicles that act as intercellular messengers and contain bioactive substances in their interior, such as DNA, RNA (miRNAs and mRNA), lipids, peptides and growth factors, which ensure the correct functioning of the surrounding microenvironment⁴.

metabolic stimulant,
skin and hair follicle
regenerator

helps to
strengthen the
skin barrier

activator of
microcirculation

*Exclusive active substance in c.prof 231 dutexome solution and tricology lotion

Cross-cutting active substances of the range



Plant extracts - **Slow down hair loss**



Serenoa repens
(Saw palmetto)



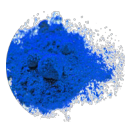
5α-BLOCKER®:

*Carthamus tinctorius, Oryza sativa,
Opuntia ficus-indica and
Piper nigrum*

5α-reductase inhibitors (block DHT)

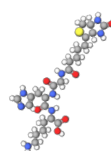


Biomimetic peptides - **Hair growth stimulators**



Copper tripeptide-1

Hair growth factor
Proliferation of DPCs
Promotes angiogenesis



Biotinoyl tripeptide-1 + oleanolic acid + apigenin

Metabolism
5α-reductase inhibitor
Increases microcirculation



Vitamins and minerals - **Nutrition and keratin synthesis**



Vit. B complex

biotin (B7), dexpanthenol (B5) and pyridoxine (B6)

Seborregulator
Anti-inflammatory
Increases microcirculation
Keratin synthesis in hair

³⁰
Zn
Zinc

³⁴
Se
Selenium

Seborregulator
Protein synthesis, antioxidant and
maintenance of hair and nails

Other active substances in the range

Dutasteride



Curcuma longa



Palmaria palmata



Melatonin



Azelaic acid



Arginine, adenosine,
taurine and caffeine



Scientific endorsement

+15 proprietary *in vitro*, *ex vivo* and cell microscopy studies⁵

proven clinical efficacy⁶

24 

markers and genes studied

2 

cell microscopy studies

1 

biodistribution study in skin explant

✓ decreased loss
✓ density and strengthening

✓ improved hair quality

In vitro studies: summary



MEDICAL/PROFESSIONAL TREATMENT

At the clinic

Topical home care

new



evo

ANTI-HAIR

Blocks DHT*

Follicular regenerator (DPCs**)

Seborregulator

Fortifying

DUTASTERIDE + PLANT EXOSOMES +
COPPER PEPTIDES + VIT. B COMPLEX

PALMARIA PALMATA + VIT. B COMPLEX + ZINC

How to use: microneedling



- Facilitates the penetration and absorption of topicalsolutions
- Physical stimulation of the hair follicle: vascularization, growth factors and neocollagenesis

How to use



- Daily use
- Apply to wet scalp and massage gently, leave on for 2-3 minutes and rinse

HOME TREATMENT

Nutraceutical



LOSS ACTION

Blocks DHT*

Follicular regenerator (DPCs**)

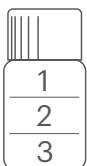
Blocks DHT*

Metabolism

SERENOA REPENS
(SAW PALMETTO) + ZINC

PLANT EXOSOMES + PEPTIDES
+ VIT. B COMPLEX

5 α -BLOCKER*
+ VIT. B COMPLEX + ZINC

How to use x3  

Distribute one vial in 3 applications per week. Apply up to the dosage line on alternate nights.

How to use 

1 capsule daily with a glass of water in the morning.

*DHT: Dihydrotestosterone **DPCs: Dermal papilla cells



actives professional solutions

contains
exosomes



pH 6.5-7.5

pol. (-)

Autoclaved solution

c.prof 231 dutexome solution

It controls hair loss by normalizing hair cycles, it inhibits DHT formation, it increases hair density and it reduces the miniaturization of hair follicles. Improves anchorage to the hair follicle. Activates microcirculation and improves the nutrient supply to the follicle.

indications:

- Hair loss due to hormonal factors in both sexes
- Reactive hair loss in the postpartum period at the end of breastfeeding
- Progressive hair loss in menopause
- Seasonal hair loss or due to nutritional deficits

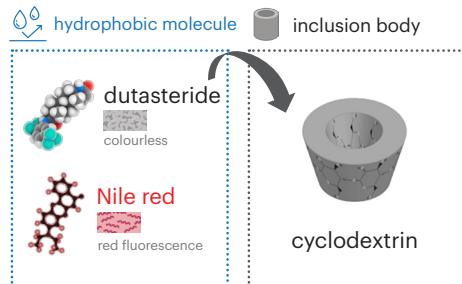
active substances:

- Dutasteride 0.05%
- Aloe vera exosomes
- Copper tripeptide-1
- Vit. B complex (biotin, dexpanthenol and pyridoxine)

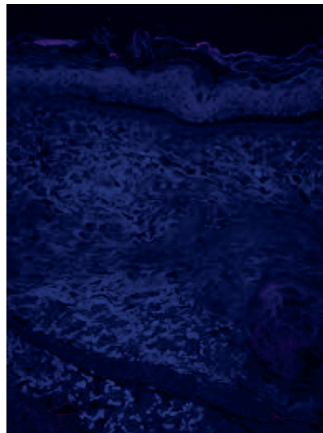
Ex vivo study

BIODISTRIBUTION STUDY IN SKIN EXPLANT¹⁰

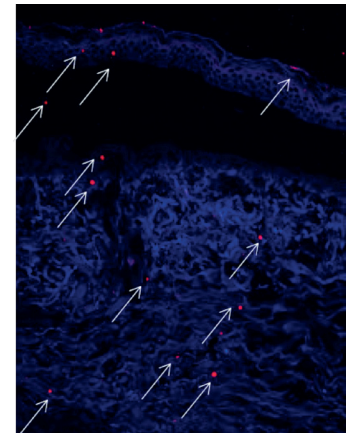
Formula solubilized using inclusion bodies, which allow greater biodistribution in the tissue. It also prevents the dutasteride from precipitating into solution over time.



Control
c.prof 231 dutexome solution



Solution A
c.prof 231 dutexome solution + Nile red

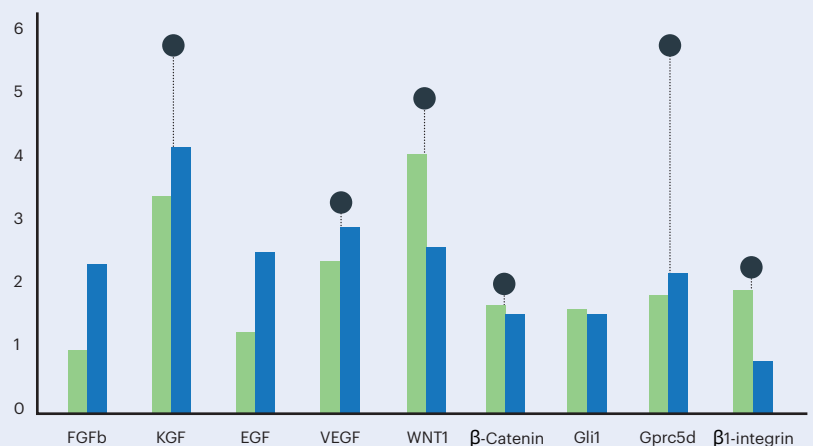
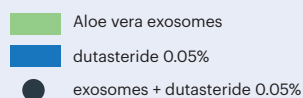


In vitro studies

ENHANCES HAIR GROWTH¹¹, due to the synergistic and complementary effect

Aloe vera exosomes + dutasteride 0.05%.

Increased genes and markers involved in the growth phase (anagen)



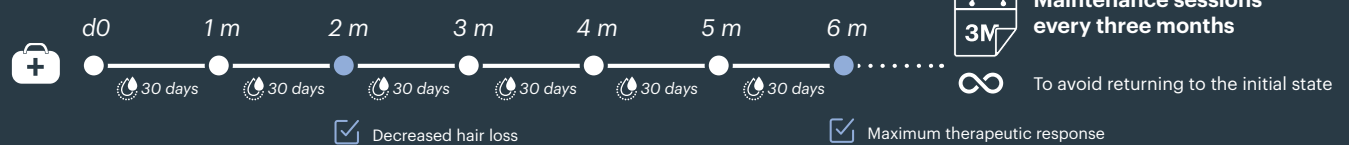
Study conducted on DPCs treated for 24h with Aloe vera exosomes, dutasteride 0.05% and the combination of both

Complete treatment protocol

c.prof + tricology treatment programme

Protocol at the clinic

7 shock sessions (6 m)



Clinic treatment phases



session duration:
15 minutes

DIAGNOSIS

- Visual inspection: combing by locks
- Trichoscopy - Trichoscope (x10-x70)



Exclusion criteria in treatment with c.prof 231 dutexome solution:

- Under 14 years of age
- Pregnant or breast-feeding women
- Men or women who would like to have a baby within the next 6 months

PREPARATION

pre-procedure cleansing solution



- degreases the scalp
- microexfoliation
- favours absorption of the active substances

DISINFECTION

70% alcohol or chlorhexidine

APPLICATION

microneedling

- **Application area:** scalp
- **Depth:** 0.25 mm
- **Speed:** 10,000 cycles/min
- **Volume of 2-2.5 ml** of c.prof 231 dutexome solution



Do not exceed the recommended quantity

Home routine



24h

Start of home treatment

6 m

End of intensive treatment at the clinic and continuation with home treatment



tricology

Topical shampoo and lotion solutions, reinforced with food supplements

Recommendations for use of the home treatments

- For 24h following the professional treatment, avoid showers, saunas or gyms. After 24 hours have elapsed, the home treatment can be started (shampoo and lotion).
- The shampoo should be used daily.
- The lotion and capsules must be continued for 4 months without interruption. It can be extended beyond the 6 months of professional treatment.
- The entire home treatment cycle can be repeated without a break.
- It is recommended that use of the capsules be discontinued during pregnancy or breast-feeding.

hair care solutions

contains
exosomes



*Sufficient quantity for 2 months of treatment.

1 vial per week
Apply 1/3 of the vial every other night

tricolony lotion

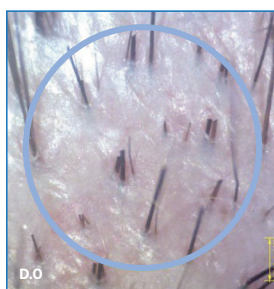
Revitalizing anti-hair loss lotion

Lotion that slows down and prevents hair loss and stimulates growth thanks to its revitalizing, redensifying and protective action on the scalp.

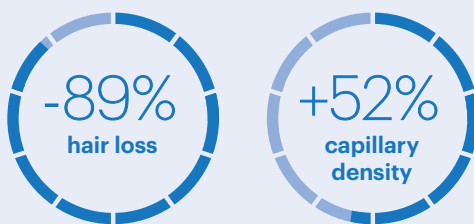
active substances:

- Aloe vera exosomes
- *Serenoa repens* (Saw palmetto)
- *Curcuma longa*
- Copper tripeptide-1
- Biotinoyl tripeptide-1
- + oleanolic acid + apigenin
- Oligopeptide-2
- Vit. B complex (biotin, pyridoxine and dexpanthenol)
- Zinc
- Azelaic acid and hyaluronic acid
- Taurine and Caffeine

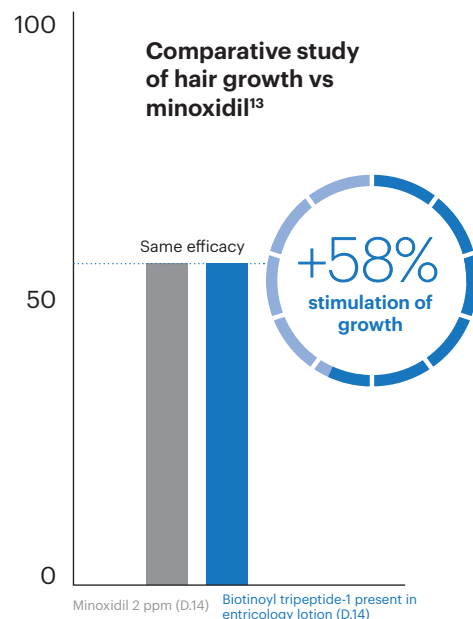
Clinical efficacy



Results at 150 days (n=40)¹²



Efficacy study of *Curcuma longa* in tricolony lotion

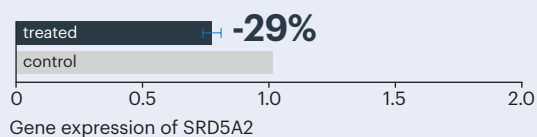


In vitro studies

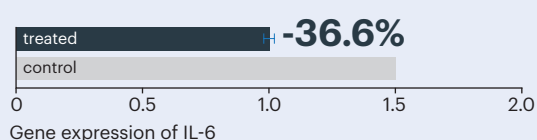
Corrective effect

Decrease of markers involved in telogen phase (hair loss)

Inhibition of the 5 α -reductase enzyme¹⁴



Anti-inflammatory¹⁵



Preventive effect

Increased expression of genes involved in the anagen phase (growth)¹⁶

Growth factors

⬆️ **FGFb, FGF1, KGF**

Activation through WNT start of the anagen phase

⬆️ **WNT1**

Vascular endothelial growth factor

⬆️ **VEGF**

Proteins via Hedgehog (Shh) capillary growth

⬆️ **Gli1, Gli2**

Culture of DPCs treated with the active substances of tricolony lotion after 24 hours: increase in markers of hair regeneration and growth.



☀️ or 🌙 Daily

tricology shampoo

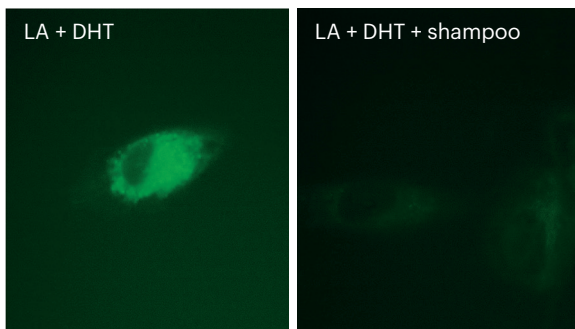
Energizing, fortifying and seborregulator shampoo

Anti-hair loss shampoo with a revitalising and sebum-regulating action that reduces hair loss. Compatible with skin pH.

active substances:

- *Serenoa repens* (Saw palmetto)
- *Palmaria palmata*
- Copper tripeptide-1
- Zinc
- Vit. B complex (biotin, pyridoxine and dexpanthenol)
- Niacinamide (B3)

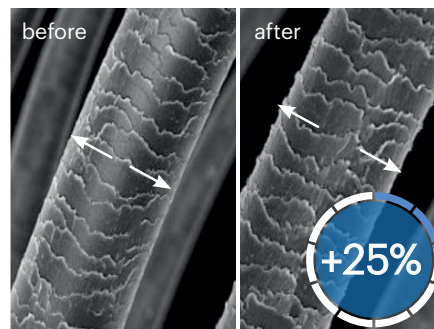
Seborregulator effect¹⁷



Sebocytes treated for 48h with sebum stimulators (linoleic acid and dihydrotestosterone) and treated with the active substances of the shampoo.

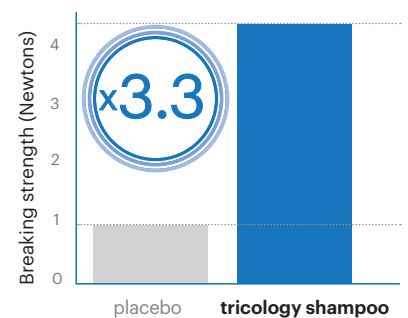
■ sebum

Increase of thickness¹⁸



Capillary fibre diameter measurement with a micrometer. ($p < 0.0001$)

Breaking strength¹⁹



Traction resistance test. Measured in Newtons. ($p < 0.0001$)



*Sufficient quantity for 2 months of treatment.

☀️ 1 capsule per day

tricology caps

Anti-hair loss, fortifying and metabolic stimulant

Daily food supplement that slows hair loss and strengthens the hair follicle

active substances:

- 5α-BLOCKER[®]: *Carthamus tinctorius*, *Oryza sativa*, *Opuntia ficus-indica* and *Piper nigrum*
- Zinc and selenium
- Vit. B complex (biotin, pyridoxine and dexpanthenol)
- Vitamins E and D3
- Cobalamin (Vit. B12)



In&Out strategy



Nourishes hair from the inside



Improves the anchoring of hair to the hair follicle



Protects against oxidative stress



Promotes the correct development of hair and nails



100% vegan dietary supplement fortified with vitamin B12

hair care solutions

Clinical studies⁶

Female androgenic alopecia (FAGA)

c.prof 231 dutexome solution + trichology home range



Female reactional hair loss

c.prof 230 hair loss solution + trichology home range



Male androgenic alopecia (MAGA)

c.prof 231 dutexome solution + trichology home range



trichoscopy



Male androgenic alopecia (MAGA)

c.prof 231 dutexome solution + trichology home range



Male androgenic alopecia (MAGA)

c.prof 231 dutexome solution + trichology home range



Male androgenic alopecia (MAGA)

c.prof 231 dutexome solution + trichology home range



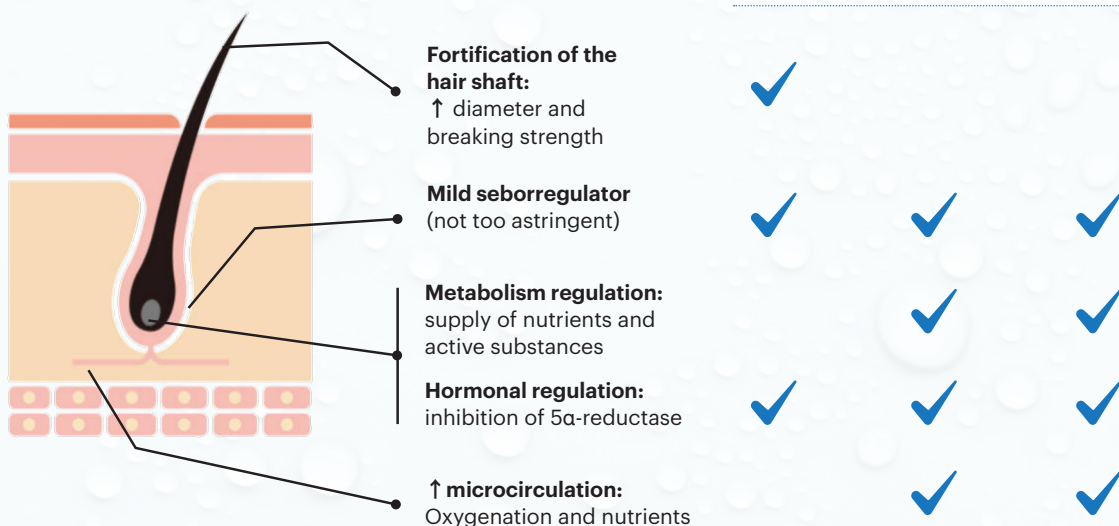
trichoscopy



hair care solutions

Multimodal approach

 Home strategy for maximizing the results of the clinical protocol:



Advanced textures and pharmaceutical forms



Convenient formats



2 months of treatment in a single pack



Suitable for all types of hair

Key data for proper hair care

Wash daily with shampoo. Apply to the **scalp**, **massage in gently** and leave on for **2-3 minutes** to increase the penetration of the active substances. Apply moisturising masks according to individual needs.

Opt for **lukewarm water in the shower** to avoid drying the hair out and finish with a **cold rinse** for **extra shine**.

Brush hair carefully when dry using a **flexible brush**, **starting at the ends** to avoid breaking the hair.

Drink 2 litres of **water** and eat a **balanced diet**, supplementing daily with **tricology caps**.

Protect yourself from **UV** radiation in the summer with **hair sunscreen**, caps or hats.

Other mesoestetic® hair solutions



Professional/medical use



5 vials x 5 ml

pH 6.0-7.5

pol. (-)

c.prof 230 hair loss solution

Revitalising action and activates hair growth.

Stops hair ageing, revitalising and strengthening the hair.

indications:

- Reactional, seasonal or nutritional deficiency hair loss in both sexes
- Improves microcirculation and nutrient supply to the follicle, revitalizing and strengthening it

active substances:

- Copper tripeptide-1
- Vit. B complex (biotin, dexpanthenol and pyridoxine)
- Zinc
- Caffeine
- Capsaicin



For medical use only

Intradermal medical solutions



mesohyal® biotin

Reactivation solution of skin and scalp cell metabolism.

Composition: HA, biotin (0.5%), other B-vitamins and mineral salts



mesohyal® NCTC 109

Revitalizing solution that moisturizes and strengthens the scalp.

Composition: HA, 20 amino acids, 12 vitamins (A, B, C, E, I), coenzymes and mineral salts

Scan the QR code to access the complete clinical study.



hair care solutions

Formula innovation.

Highly effective active ingredients and plant exosomes to strengthen and control hair loss.

Multimodal approach.

Professional/medical treatment enhanced with home adjuvants for a comprehensive approach.

Scientific endorsement.

In vitro, *ex vivo* and clinical studies that demonstrate the efficacy of the range.

Innovative professional/medical solution accompanied by a trichology range of evolved and reinforced home care products.



1. Kamp E, Ashraf M, Musbahi E, DeGiovanni C. Menopause, skin and common dermatoses. Part 1: hair disorders. Clin Exp Dermatol. 2022 Dec;47(12):2110-2116. **2.** Khashouf Maksoud K. Revisión actualizada del tratamiento de la alopecia androgénica femenina. Medicina Estética. 2022;71(2):9-20. **3.** Piraccini BM, Alessandrini A. Androgenetic alopecia. G Ital Dermatol Venereol. 2014 Feb;149(1):15-24. **4.** Casado-Díaz A, Quesada-Gómez JM, Dorado G. Extracellular Vesicles Derived From Mesenchymal Stem Cells (MSC) in Regenerative Medicine: Applications in Skin Wound Healing. Bioeng Biotechnol. 2020 Mar 3;8:146. **5.** *In vitro* and *ex vivo* efficacy studies of c.prof 231 dutexome solution and the trichology range. 2024. Biotechnology unit. mesoestetic® Pharma Group S.L. **6.** Clinical safety and efficacy study of c.prof 231 dutexome solution and the trichology home range in 60 patients with androgenic alopecia. 2024 Medical Affairs Unit. mesoestetic® Pharma Group S.L. **7.** Kumar N, Rungsevivijitprapa W, Narkkhong NA, Suttajit M, Chaiyasut C. 5α-reductase inhibition and hair growth promotion of some Thai plants traditionally used for hair treatment. J Ethnopharmacol. 2012 Feb 15;139(3):765-71. **8.** Cestone, E., Sparta, E., Nobile, V., Zaccaria, V., Pappalardo, F., Insolia, V., Malfa, G. A., Buonocore, D., & Dossena, M. (2023). Efficacy of *Oryza sativa* L. (Black Rice) and *Opuntia ficus-indica* L. Blend in Men with Androgenetic Alopecia: A Randomized, Double-Blind, Placebo-Controlled Clinical Trial. Journal of Cosmetology & Trichology, 9(3). **9.** Jonas A, Rosenblatt G, Krapf D, Bitterman W, Neeman I. Cactus flower extracts may prove beneficial in benign prostatic hyperplasia due to inhibition of 5α reductase activity, aromatase activity and lipid peroxidation. Urol Res. 1998;26(4):265-70. **10.** *Ex vivo* study of skin explant biodistribution of dutasteride from c.prof 231 dutexome solution. 2024. Biotechnology unit. mesoestetic® Pharma Group S.L. **11.** *In vitro* study of increased gene expression of markers involved in the anagen phase in DPCs treated for 24h with Aloe vera exosomes, dutasteride 0.05% and the combination of both. 2024. Biotechnology unit. mesoestetic® Pharma Group S.L. **12.** Double-blind clinical study on efficacy in a group of 40 volunteers with alopecia of different aetiologies after the topical use of *Curcuma longa*, contained in trichology lotion, after 150 days. May 2016 **13.** Comparative study on hair shaft growth in the presence of 2 ppm Biotinoyl Tripeptide-1 vs 2 ppm minoxidil (10 µM) Sederma 11/16 PCS TB 022 V5 EN. **14.** *In vitro* study of SRD5A2 gene expression inhibition in DPCs treated for 24h with trichology lotion. 2024. Biotechnology unit. mesoestetic® Pharma Group S.L. **15.** *In vitro* study of the decrease of IL-6 gene expression in DPCs treated after 72h with trichology lotion after inducing damage with DHT. 2024 Biotechnology unit. mesoestetic® Pharma Group S.L. **16.** *In vitro* study of increased gene expression of markers involved in the anagen phase of DPCs treated for 24h with trichology lotion. 2024. Biotechnology unit. mesoestetic® Pharma Group S.L. **17.** Cell microscopy study of sebum reduction in sebocytes treated after 48h with trichology shampoo after inducing sebum production with linoleic acid and DHT. Biotechnology unit. mesoestetic® Pharma Group S.L. **2024** **18.** *Ex vivo* efficacy study of hair fibre diameter measurement with a micrometer after topical use of *Palmaria palmata*, contained in trichology shampoo vs a placebo. **19.** *Ex vivo* efficacy study of average tensile strength test in Newtons after topical use of *Palmaria palmata*, contained in trichology shampoo.



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